

Carven Maceke

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Objective

Junior Software Engineer and IT graduate with hands-on experience in backend development, RESTful API development, Java, Spring Boot, SQL, database integration, and web application development. Experienced in developing backend services, writing SQL queries, debugging applications, testing software, and collaborating using Git/GitHub. Recently completed structured AWS cloud training and earned the AWS Certified Cloud Practitioner certification, with knowledge of cloud computing, AWS services, security, networking, and deployment concepts. Currently working as a Software Development Intern, contributing to a full-stack system that integrates web development, AI/ML, and IoT components, and applying software engineering practices to real-world development projects. Seeking a Junior Software Engineer, Software Developer, Backend Developer, or Full Stack Developer opportunity to contribute to reliable and scalable software solutions.

Education

Nelson Mandela University

Port Elizabeth

National Diploma in Information Technology in Software Development - NQF Level 6 — Completed (2024)

Experience

Forge Academy x Nokia

Clearwater Office Park, Millennium Boulevard, Roodepoort

Software Development Internship

2026-03 – Current

- Contributing to a large-scale internal system combining full-stack web development, AI/ML services, and IoT integration, developed using Java, Python, Spring Boot, and cloud technologies.
- Developing backend services using Spring Boot, Spring MVC, Spring Data JPA, and Hibernate, exposing RESTful APIs consumed by both web and IoT clients.
- Building front-end components using Bootstrap and JavaScript, including responsive layouts and data visualisation dashboards.
- Applying Python for AI/ML functionality within the system, including predictive modelling and computer vision using Scikit-learn, OpenCV, and PyTorch/TensorFlow.
- Integrating ESP32-based IoT devices using MQTT for real-time sensor monitoring as part of the system's IoT architecture.
- Using MySQL/PostgreSQL for relational database design, and Docker, Git/GitHub, and CI/CD pipelines to support development, testing, and deployment.
- Applying REST, MVC, layered, IoT, and AI-service integration architecture patterns across the project.
- Completed AWS cloud computing training covering core cloud concepts, AWS services, security, networking, and deployment fundamentals; earned the **AWS Certified Cloud Practitioner** certification.
- Applying problem-solving, troubleshooting, and technical skills within a professional software development environment.

Skills Used: Java • Python • JavaScript • Spring Boot • Spring MVC • Spring Data JPA • Hibernate • REST APIs • Bootstrap • Data Visualisation • Scikit-learn • OpenCV • PyTorch/TensorFlow • Predictive Modelling • Computer Vision • MySQL • PostgreSQL • ESP32 • MQTT • IoT Integration • AWS • Docker • Git/GitHub • CI/CD • Linux • Troubleshooting • Problem Solving

K & K Media & Solwazi Institute

Constantia Square Office Park, Midrand

IT Software, Systems & Support Intern

2025-09 – 2026-03

- Supported software and web systems, including updating and maintaining a WordPress-based platform.
- Conducted interviews and mentored/supported interns, assisting with their onboarding and technical development.
- Set up and configured email accounts and systems for staff, alongside broader IT system configuration.
- Analysed and cleaned messy, unstructured datasets in Excel, improving data accuracy, structure, and usability for reporting.
- Diagnosed and resolved Microsoft Office application issues, improving software reliability for end users.
- Troubleshoot technical issues across hardware and software, improving system reliability.

- Configured IT systems (email, networks, printers), strengthening system integration understanding.
- Collaborated with team members to resolve issues and improve operational workflows.

Skills Used: Data Analysis • Data Cleaning • Troubleshooting • WordPress • IT Support • Networking • Interviewing • Mentoring & Coaching • Email Configuration • Microsoft Office Support • Problem Solving

ZAIO Institute of Technology

96 Rivonia Rd, Dennehof, Sandton

Junior Java Developer

2023-09 – 2024-08

- Developed backend services and RESTful APIs using Java and Spring Boot
- Designed and optimized SQL queries for data processing and reporting
- Participated in debugging and resolving application issues in a team environment
- Assisted in testing, deployment, and configuration of applications
- Used Git/GitHub for version control and collaborated on shared codebases
- Created technical documentation to support maintainability and knowledge sharing

Key Skills: Java • Spring Boot • SQL • API Development • Debugging • Git/GitHub • Testing

Projects

Portfolio Website

- Designed and developed a responsive web portfolio using HTML, CSS, and JavaScript.
- Implemented interactive user interface components and responsive layouts to improve usability across different devices.
- Deployed the website using Netlify, gaining practical experience in web deployment and hosting.

Link: <https://carven.netlify.app>

Skills Used: HTML • CSS • JavaScript • Web Development • Responsive Design • UI/UX • Web Deployment

Healthcare System Web Application

Nelson Mandela University | 2023-03 – 2023-11

- Developed a responsive healthcare web application using HTML, CSS, JavaScript, and MySQL.
- Designed and integrated a MySQL database for secure data storage and retrieval.
- Applied the MVC architectural pattern to create a structured and maintainable application.
- Used Git/GitHub for version control and team collaboration during the development lifecycle.
- Participated in application testing, debugging, technical documentation, and project presentations.

Skills Used: HTML • CSS • JavaScript • MySQL • MVC • Git/GitHub • Database Integration • Web Development • Software Testing • Debugging • Documentation • Team Collaboration

Technical Skills

Languages

Java • Python • JavaScript • SQL • HTML5 • CSS3 • C# (Basic)

Backend

Spring Boot • Spring MVC • Spring Data JPA • Hibernate • REST APIs • Backend Development • API Integration • ASP.NET MVC

Frontend

Bootstrap • JavaScript • Responsive Web Development • Data Visualisation • HTML • CSS

AI/ML

Python • Scikit-learn • OpenCV • PyTorch/TensorFlow • Predictive Modelling • Computer Vision

Database

MySQL • PostgreSQL • Microsoft SQL Server • SQL • Relational Database Design • Query Optimization • Database Integration

IoT

ESP32 • MQTT • Sensor Integration • Real-Time Monitoring

Cloud & DevOps

AWS Cloud • AWS Certified Cloud Practitioner • EC2 • S3 • IAM • VPC • CloudWatch • Docker • Cloud Computing • Cloud Security Fundamentals • Networking • Deployment Concepts • CI/CD • Linux

Architecture

REST • MVC • Layered Architecture • IoT Architecture • AI-Service Integration • Design Patterns

Development Tools

Git • GitHub • Version Control • Debugging • Software Testing • Technical Documentation

IT & Systems

Hardware & Software Troubleshooting • IT Support • Network Configuration • System Configuration • Email Configuration • Microsoft Office Support • Technical Support

Professional Skills

Problem Solving • Team Collaboration • Interviewing • Mentoring & Coaching • Adaptability • Attention to Detail • Time Management • Communication • Continuous Learning

Certifications

- **AWS Certified Cloud Practitioner** – Amazon Web Services (2026)
- **Java Certification** – ZAIO Institute of Technology (December 2023)
- **SQL Certification** – ZAIO Institute of Technology (December 2023)
- **Quality Work Experience** – YES4Youth (September 2024)
- **Cybersecurity for Everyone** – University of Maryland (February 2024)
- **CCNAv7: Introduction to Networks** – Cisco Networking Academy
- **Introduction to Cybersecurity** – Cisco Networking Academy

References

Available upon request